

SUBJECT CODE NO: - RNP-8003
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B]Mech, Civil, Comp. AI & ML,EEE
Examination January/February 2025
Engineering Chemistry

[Max. Marks: 60]

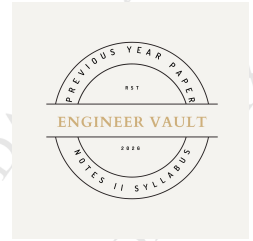
N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Que. nos 2, 3, 4, 5, 6 and 7

Q1 Solve the following questions (any 10)

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- 1) Total hardness of water is 1250 ppm temporary hardness of water is 350 ppm calculate permanent hardness of water.
 - a) 900ppm
 - b) 610 ppm
 - c) 1250 ppm
 - d) 350 ppm
- 2) In Carbon-Carbon (C-C) composites, what is the role of the carbon fiber reinforcement?
 - a) To increase the material's thermal conductivity
 - b) To provide strength, stiffness, and fracture toughness
 - c) To enhance the matrix's electrical conductivity
 - d) To decrease the weight of the composite material
- 3) A 100 mL water sample contains 50 mg/L of Ca^{2+} and 30 mg/L of Mg^{2+} ions. Calculate the total hardness of the water in mg/L as CaCO_3 .
 - a) 80 mg/L
 - b) 100 mg/L
 - c) 120 mg/L
 - d) 150 mg/L
- 4) Which of the following best describes the storage medium used in Computed Radiography (CR)?
 - a) Digital storage devices like hard drives or cloud storage
 - b) Photo Stimulable phosphor (PSP) plates that are later scanned
 - c) Amorphous selenium detectors
 - d) Optical storage films
- 5) What is the main difference between ultrasonic testing and x-ray testing?
 - a) UT uses electromagnetic waves, while x-ray testing uses sound waves
 - b) UT requires direct contact with the material, while x-ray testing can be non-contact
 - c) UT is used for testing electrical properties, while x-ray testing is used for testing mechanical properties
 - d) UT is only used for metals, while x-ray testing is for all materials
- 6) What is the main difference between Computed Radiography (CR) and Digital Radiography (DR)?
 - a) CR uses phosphor plates, while DR uses a direct digital detector
 - b) DR uses phosphor plates, while CR uses a direct digital detector
 - c) CR uses film, while DR uses a digital sensor
 - d) There is no difference between CR and DR



- 7) What is the primary cathodic reaction that occurs during wet corrosion?
- $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$
 - $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
 - $\text{Fe} + 2\text{H}_2\text{O} \rightarrow \text{Fe}(\text{OH})_2 + 2\text{H}^+$
 - $\text{Fe}(\text{OH})_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{H}_2\text{O}$
- 8) What is the difference between Gross Calorific Value (GCV) and Net Calorific Value (NCV)?
- GCV includes the latent heat of water vapour, while NCV does not.
 - GCV and NCV are the same values.
 - NCV includes the latent heat of water vapour, while GCV does not.
 - GCV is always higher than NCV by a fixed amount.
- 9) In laser cladding, the laser beam is typically used to:
- Heat the material to a molten state
 - Solidify the material instantly
 - Add a layer of coating material directly to the part surface
 - Remove any impurities from the base material
- 10) Which of the following is the primary difference between flash point and fire point?
- Flash point is the temperature at which the liquid begins to burn, while fire point is the temperature at which the liquid's vapour ignites.
 - Flash point is a lower temperature than fire point, and fire point is the temperature at which combustion can be sustained.
 - Flash point is determined in a closed system, while fire point is determined in an open system.
 - There is no difference between flash point and fire point.
- 11) What is the composition of LNG in terms of major components?
- 80-90% methane, 5-10% ethane, 1-5% propane, and trace amounts of other gases
 - 70-80% methane, 10-20% ethane, and 5-10% butane
 - 50-60% methane, 20-30% ethane, and 10-20% propane
 - 90-95% propane, 5-10% methane, and trace amounts of ethane
- 12) What is the primary principle behind the operation of bio electrochemical batteries (BEBs)?
- They generate electricity using only chemical reactions between metals.
 - They use living microorganisms or enzymes to catalyze redox reactions and generate electrical current.
 - They operate based on heat from the environment.
 - They use radioactive materials to produce electrical energy.
- 13) Viscosity at 100°C of the sample oil (U) = 18 cSt
 Viscosity at 100°C of the standard oil with a low VI (L) = 12 cSt
 Viscosity at 100°C of the standard oil with a high VI (H) = 24 cSt
 Calculate the Viscosity Index (VI).
- 50
 - 60
 - 56
 - 70

SECTION B

- Q2** a) Give synthesis properties and applications of Thermosetting polymer **05**
b) Give properties and applications of CNT **05**
- Q3** a) writes a note on electrochemical corrosion **05**
b) Explain corrosion testing methods **05**
- Q4** a) Explain composition, properties, and applications of CNG **05**
b) Writes a note on alkaline fuel cell **05**
- Q5** a) Explain synthesis properties and applications of PVA **05**
b) Calculate total hardness of water containing following impurities **05**
- | Impurity | Wt | Molecular wt. |
|------------------------------------|---------|---------------|
| CaCO ₃ | 13.5ppm | 100 |
| MgSO ₄ | 65.2ppm | 120 |
| Ca(HCO ₃) ₂ | 21ppm | 162 |
- Q6** a) write a note on electroplating **05**
b) Explain casting, forging, rolling, machining **05**
- Q7** a) Define fuel give its classification and its unit **05**
b) Explain proximate analysis **05**

